

iMHEA Data Processor — Installation Manual

Version 1.0 • July 2026

Table of Contents

About this software

The **iMHEA Data Processor** processes rainfall and streamflow data from hydrological monitoring catchments following the methodology of the Regional Initiative for Hydrological Monitoring of Andean Ecosystems (iMHEA), published in Ochoa-Tocachi et al. (2018), *Scientific Data* 5:180080.

It is distributed in three forms — install whichever suits you:

Component	For	Requires
Desktop application	Field teams and analysts; no programming	Nothing (standalone)
Command-line tool (imhea)	Batch processing, automation	Python 3.10+
Python library (import imhea)	Researchers, custom analyses	Python 3.10+

Option A — Desktop application (recommended)

A.1 From a release package

If you received (or downloaded from the project's GitHub Releases page) a ZIP file for your operating system:

Windows

1. Unzip iMHEA-Data-Processor-windows.zip anywhere (e.g. Documents).
2. Open the folder and double-click iMHEA-Data-Processor.exe.
3. First launch only: Windows SmartScreen may warn about an unrecognised app. Click **More info** → **Run anyway**. This happens because the executable is not code-signed; it does not indicate a problem.

macOS

1. Unzip and drag iMHEA Data Processor.app to Applications (optional).

2. First launch only: **right-click the app** → **Open** → **Open**. macOS Gatekeeper blocks double-clicking unsigned apps the first time. If it still refuses, run in Terminal: `xattr -dr com.apple.quarantine "/Applications/iMHEA Data Processor.app"`

Linux

1. Unzip, then run `./iMHEA-Data-Processor/iMHEA-Data-Processor`.
2. If it reports a missing Qt platform plugin, install the system libraries: `sudo apt install libegl1 libxkbcommon0`.

A.2 Building the application yourself

If no release package is available for your platform, build it with one double-click. You need **Python 3.10 or newer** installed first (<https://www.python.org/downloads/> — on Windows, tick “Add python.exe to PATH” during installation).

1. Obtain the imhea-python folder (from the repository or from a colleague).
2. Open the packaging subfolder.
3. **macOS**: double-click `build_mac.command` (first time: right-click → Open). **Windows**: double-click `build_windows.bat`.
4. Wait a few minutes. The finished app appears in `imhea-python/dist/` and the folder opens automatically. The app is self-contained — copy or zip it to share with colleagues, who do **not** need Python.

A.3 Automated builds with GitHub Actions

If the repository is hosted on GitHub, every tagged release (e.g. v1.0.0) automatically builds and publishes packages for Windows, macOS (Intel and Apple Silicon) and Linux. Maintainers can also trigger a build manually from the **Actions** tab (*Build desktop apps* workflow); the packages appear as downloadable artifacts on the workflow run page.

Option B — Python package (CLI + library)

For users comfortable with a terminal:

```
pip install imhea          # from PyPI, when published
# or, from a repository checkout:
pip install -e ".[gui]"    # library + CLI + desktop app
```

This installs two commands:

- `imhea` — the command-line interface (see the User Manual, section 9).
- `imhea-gui` — launches the desktop application.

To run the GUI from a source checkout without installing: `python gui/run_gui.py`.

System requirements

	Minimum
Operating system	Windows 10+, macOS 12+, or Linux (glibc 2.31+)
Memory	4 GB (8 GB recommended for multi-year 5-min records)
Disk	500 MB for the application; data storage as needed
Python (options B / A.2 only)	3.10 or newer

Updating

- **Desktop app:** download or build the new version and replace the old folder/app. Project files (.imhea) are forward-compatible.
- **Python package:** `pip install -U imhea`.

Troubleshooting

“Python not found” when building. Install Python from python.org and, on Windows, re-run the installer choosing *Modify* if you forgot to tick “Add python.exe to PATH”.

The build script closes immediately (Windows). Right-click → *Run as administrator* is not needed; instead open a Command Prompt in the packaging folder and run `build_windows.bat` to see the error message.

macOS: “app is damaged or incomplete”. The quarantine flag survives some transfers. Run: `xattr -dr com.apple.quarantine "imHEA Data Processor.app"`.

The app starts but figures are blank (Linux). Set the environment variable `QT_QPA_PLATFORM=xcb` (Wayland systems) or install `libegl1`.

Cloud-synced folders. Building inside OneDrive/Dropbox/Drive folders works but syncs hundreds of megabytes; prefer building in a local folder, or pause syncing during the build.

Getting help

Report problems on the project’s GitHub *Issues* page, attaching the message shown in the app’s **Log** tab or the terminal output. Include your operating system and the software version (shown by `imhea --version` or in the app title bar).